https://k8s-school.fr

Conteneurs: l’exemple de Docker

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Plan

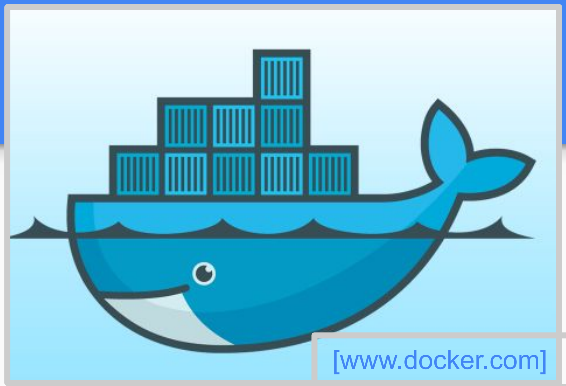
- What is Docker?
  - Docker vs. Virtual Machine
  - History, Status, Run Platforms
  - Hello World
- Images and Containers
- Volume Mounting, Port Publishing, Linking
- Around Docker, Docker Use Cases
- Hands-On Workshop

Qu’est ce que Docker?

Docker is an open-source project that automates the deployment of applications inside software containers, by providing an additional layer of abstraction and automation of operating system–level virtualization on Linux.

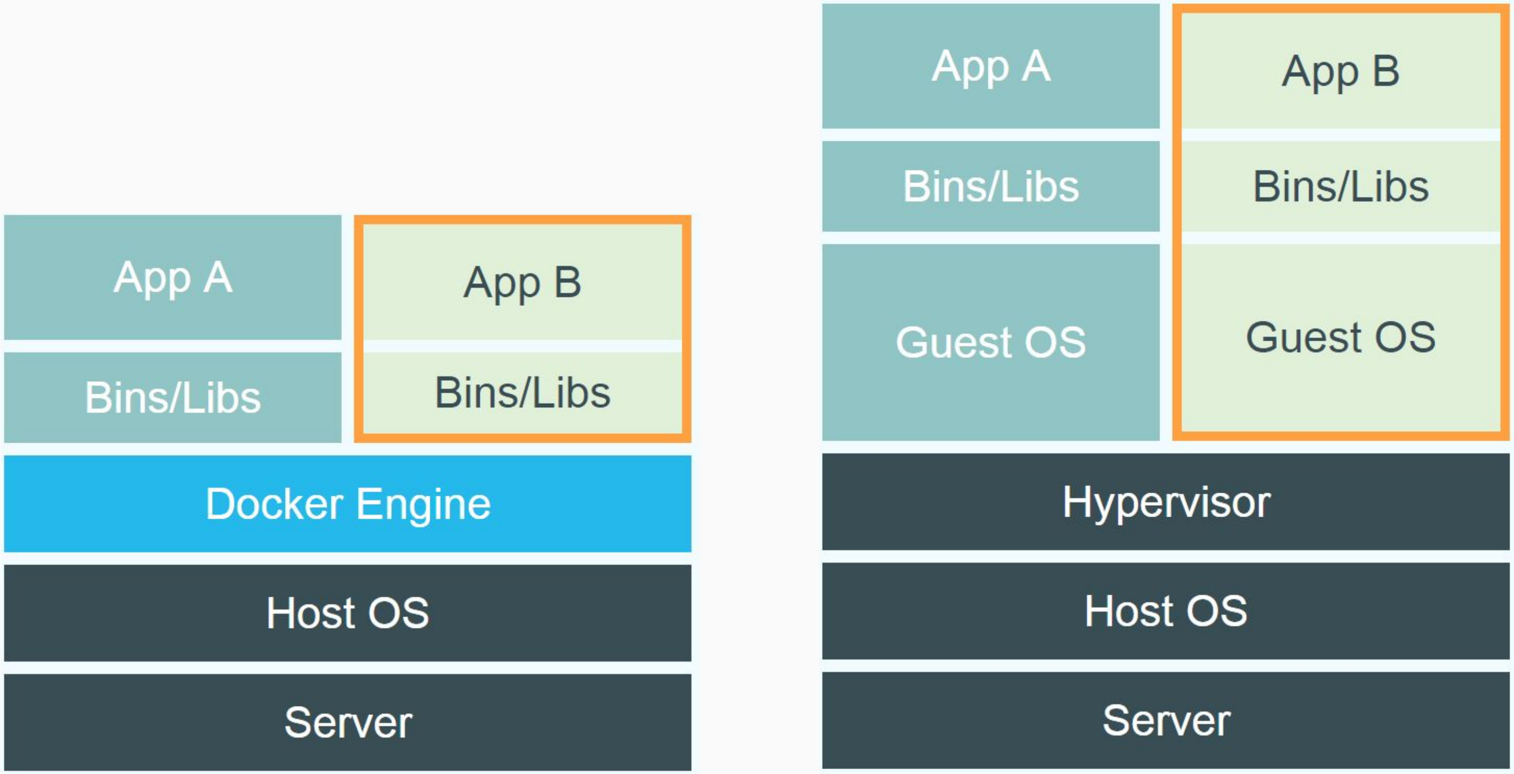
[Source: en.wikipedia.org]

Docker: Name



- Provide a uniformed wrapper around a software package: *«Build, Ship and Run Any App, Anywhere»* [www.docker.com]
  - Similar to shipping containers: The container is always the same, regardless of the contents and thus fits on all trucks, cranes, ships, ...

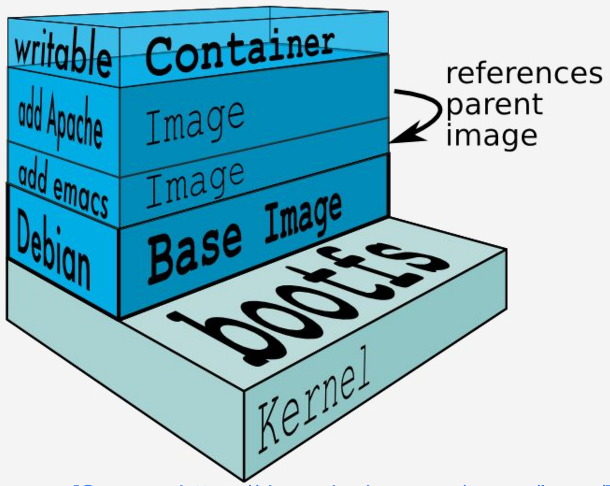
Docker vs. Virtual Machine



Source: https://www.docker.com/whatisdocker/

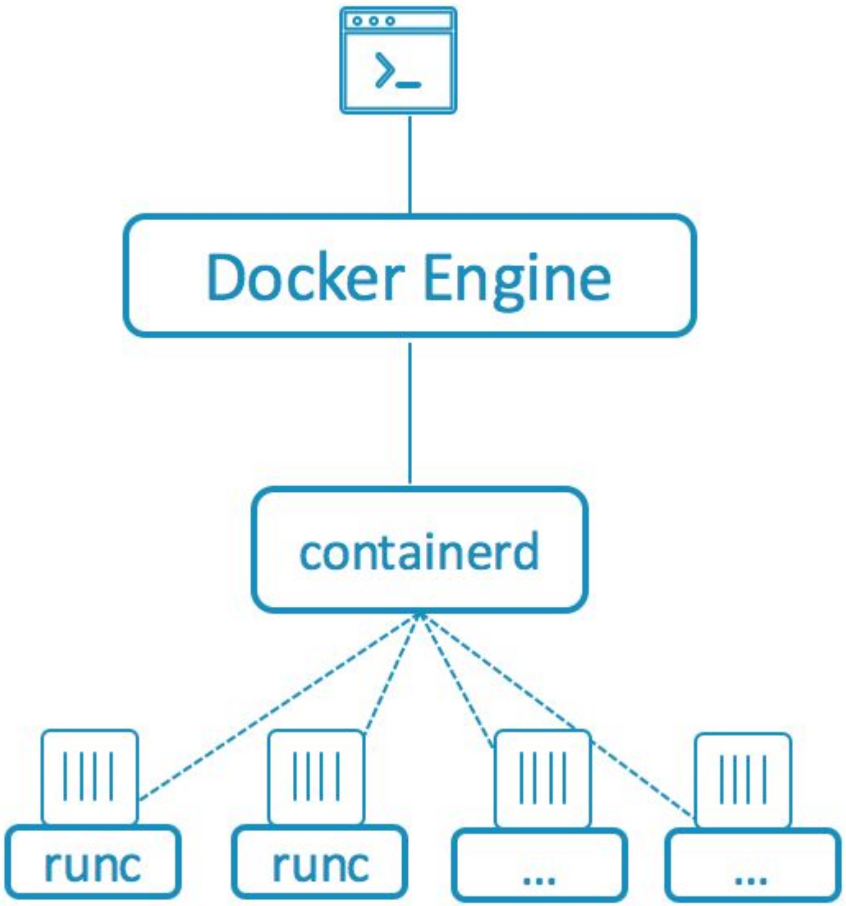
Docker Technology

- runC: The universal container runtime: **https://runc.io/**
- Open Containers Initiative: https://www.opencontainers.org/
- Layered File System https://docs.docker.com/engine/userguide/storagedriver/imagesandcontainers/



[Source: https://docs.docker.com/terms/layer/]

Modular architecture



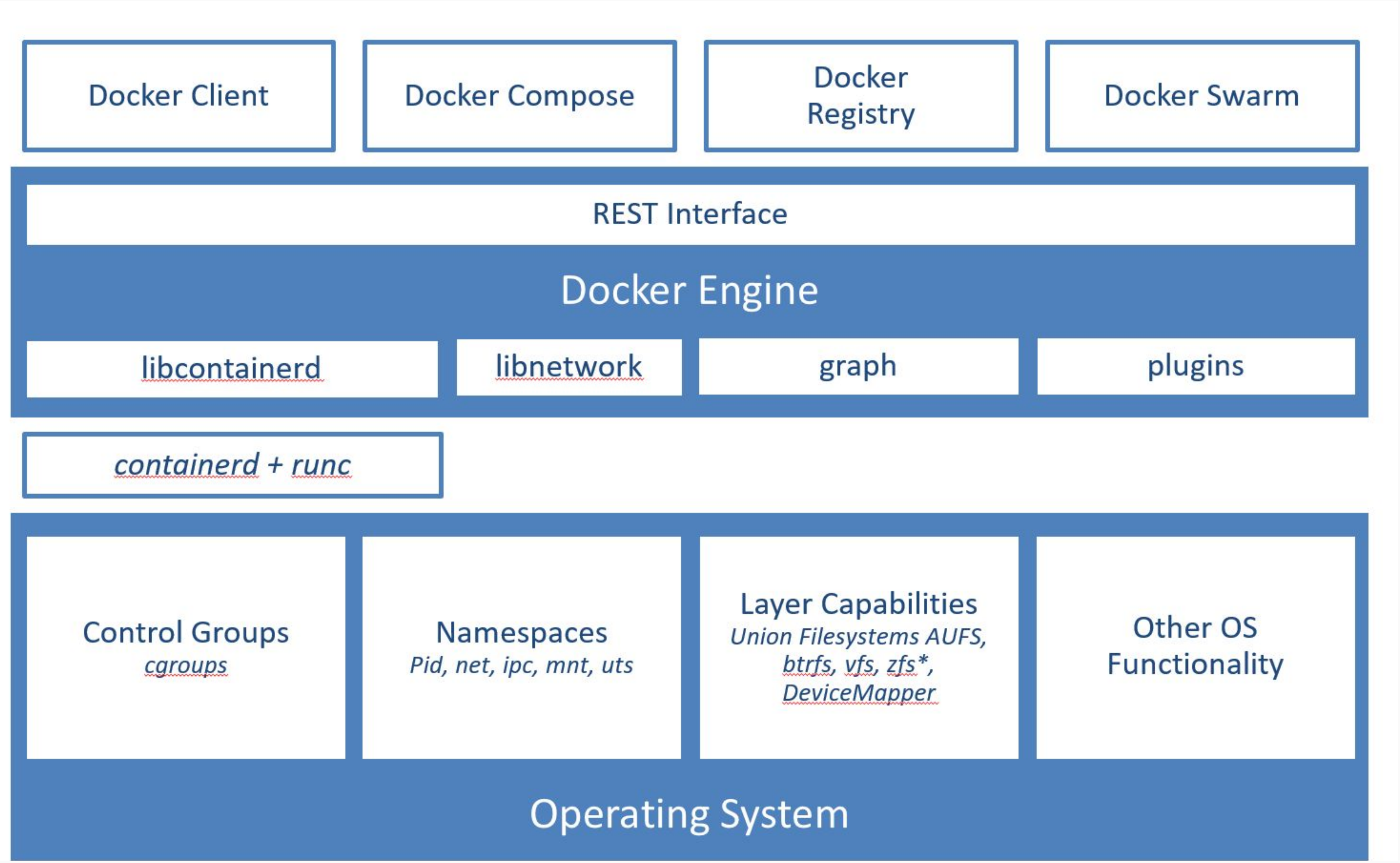
Same Docker UI and commands

User interacts with the Docker Engine

Engine communicates with containerd

containerd spins up runc or other OCI compliant runtime to run containers

Architecture in Linux



Run Platforms

- Various Linux distributions (Ubuntu, Fedora, RHEL, Centos, openSUSE, ...)
- Cloud (Amazon EC2, Google Compute Engine, Microsoft Azure)

openstack
CLOUD SOFTWARE

alpine Linux

amazon web services

Microsoft Azure

CoreOS

Hello world

Simple Command - Ad-Hoc Container

- `docker run ubuntu echo Hello World`
  - `docker images [-a]`
  - `docker ps -a`

Terminology - Image

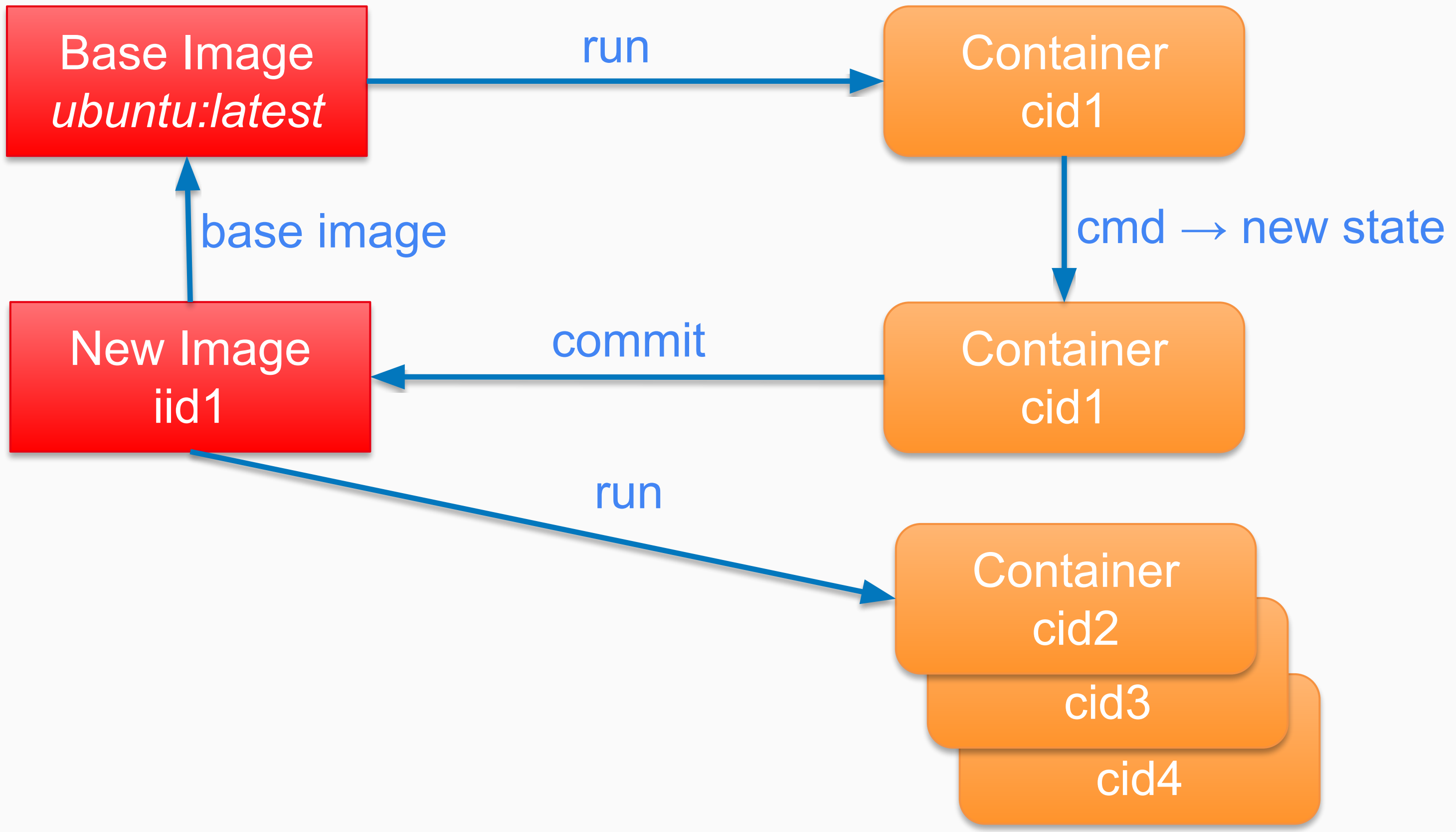
- Persisted snapshot that can be run
  - *images:* List all local images
  - *run*: Create a container from an image and execute a command in it
  - *tag*: Tag an image
  - *pull*: Download image from repository
  - *rmi*: Delete a local image
    - This will also remove intermediate images if no longer used

Terminology - Container

Runnable instance of an image

- *ps:* List all running containers
- *ps –a*: List all containers (incl. stopped)
- *top*: Display processes of a container
- *start*: Start a stopped container
- *stop*: Stop a running container
- *pause*: Pause all processes within a container
- *rm*: Delete a container
- *commit*: Create an image from a container

Image vs. Container



Dockerfile

- Create images automatically using a build script: «Dockerfile»
- Can be versioned in a version control system like Git or SVN, along with all dependencies
- Docker Hub can automatically build images based on dockerfiles on Github

Dockerfile Example

- Dockerfile:

```
FROM ubuntu
ENV DOCK_MESSAGE Hello My World
ADD dir /files
CMD ["bash", "/files/some-script.sh"]
```

- `docker build [DockerFileDir]`
- `docker inspect [imageId]`

Mount Volumes

- `docker run -ti` **`-v /hostLog:/log`** `ubuntu`

- *Run second container: Volume can be shared*

  `docker run -ti --volumes-from 1stContainerName ubuntu`

Publish Port

- `docker run -t -p 8080:80 ubuntu netcat -l -p 80`
  - Map container port 80 to host port 8080
  - Check on host: netcat localhost 8080
- *Link with other docker container*
  - *`docker run -ti --link containerName:alias ubuntu`*
  - *See link info with* `set`

Around Docker

- Docker Images: Docker Hub
- Vagrant: «Docker for VMs»
- Automated Setup
  - Puppet, Chef, Ansible, ...
- Docker Ecosystem
  - kubernetes, swarm, nomad
  - compose

Docker Hub

- Public repository of Docker images
  - https://hub.docker.com/
  - docker search [term]
- Automated: Has been automatically built from Dockerfile
  - Source for build is available on GitHub

Resource Usage

- docker top [containerId]
- docker free -m [containerId]
- docker ps

- Start 100 WebServer containers

    docker run -d -p **$hostPort**:5000 training/webapp

Docker Use Cases

- Development Environment
- Environments for Integration Tests
- Quick evaluation of software
- Microservices
- Multi-Tenancy
- Unified execution environment:

    dev → test → prod (local, VM, cloud, ...)